

Benchmarking Tabular Reinforcement Learning on Stochastic FrozenLake

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Abstract

This project extends an initial 4×4 Q-learning implementation into a stochastic FrozenLake benchmark. We focus on the slippery setting, where the executed action may deviate from the intended action, because this setting tests reinforcement learning under transition uncertainty rather than deterministic maze following. The benchmark compares classical tabular control methods including Q-learning, SARSA, Expected SARSA, SARSA(λ), Q(λ), Dyna-Q, and optimistic Q-learning, with Value Iteration as a known-model oracle. A central part of the project is the evolution of the experimental design. We first used Gymnasium as a reference point: the standard Gymnasium 8×8 map is a strong benchmark and sanity check, while Gymnasium-generated low- p random maps revealed a problem—many are connected but nearly impossible under slippery dynamics. We therefore introduced controlled maps: standard 4×4 , designed 4×4 maps with exactly four holes, fixed 8×8 , and controlled random 8×8 maps. These maps are solvable, non-trivial, reproducible, and better suited for comparing algorithms. On the controlled random 8×8 benchmark, optimistic Q-learning reaches 0.988 mean final success, followed by Expected SARSA and SARSA above 0.89. The Gymnasium standard 8×8 result is retained as an important standard comparison, and the Gymnasium random-map results are retained as a diagnostic motivation for careful map design.

1 Introduction

FrozenLake is a compact reinforcement learning (RL) benchmark in which an agent starts on a grid, avoids holes, and tries to reach a goal. Rewards are sparse: the agent receives reward 1 only at the goal and 0 elsewhere. In the slippery version, an intended move can slip to a neighboring direction, so even an optimal policy cannot be interpreted as a deterministic path-following solution.

The original project implemented Q-learning on 4×4 FrozenLake. That was a useful baseline, but it left three open questions: whether other tabular RL algorithms perform differently, whether results hold beyond a single small map, and whether hyperparameter choices are credible. This report addresses those questions with a stochastic-only benchmark, controlled map design, and focused ablation studies. The final benchmark was not chosen blindly: it was shaped by the contrast between the useful Gymnasium standard 8×8 benchmark and the often near-impossible Gymnasium low- p random maps.

The research questions are:

1. Which tabular RL methods are strongest on slippery FrozenLake?
2. Why is controlled map design necessary for fair algorithm comparison on stochastic FrozenLake?
3. Is performance mainly limited by exploration or reward propagation?
4. Does automated hyperparameter search improve the strongest methods?

2 Related Work

Q-learning is a classical off-policy temporal-difference control method that uses a max next-state target [1]. SARSA and Expected SARSA are on-policy alternatives that update values under the behavior policy rather than an ideal greedy target [2]. Eligibility-trace methods such as SARSA(λ) and Q(λ) propagate TD errors through recent state-action trajectories, which can help sparse rewards reach earlier decisions [2]. Dyna-Q combines model-free learning with planning updates from a learned transition model, reusing scarce experience more efficiently [2]. Double Q-learning reduces overestimation bias in noisy value estimates [3], though this conservatism can slow discovery when rewards are rare.

FrozenLake is widely used in Gymnasium teaching material for Q-learning and map-size sensitivity [4]. Our study follows the same qualitative motivation but adds a broader tabular comparison, oracle references, Optuna tuning, and controlled random-map tests. For tuning, we use Optuna’s Tree-structured Parzen Estimator (TPE) sampler [5]. Larger distributed sweeps could use Ray Tune with ASHA/HyperBand [6], but Optuna is lightweight and appropriate for the current coursework-scale experiments.

3 Methodology

3.1 Experimental Design Evolution

The experiment was developed in two stages. We first used Gymnasium as the reference implementation. The standard Gymnasium 8×8 map is valuable because it is widely recognized, reproducible, and neither trivial nor impossible under `slippery=True`. It therefore serves as an important standard benchmark and sanity check for the training pipeline.

The problem appeared when we moved from the standard Gymnasium map to library-generated random maps. Gymnasium’s random-map generator guarantees that a path exists from start to goal, but in slippery FrozenLake deterministic reachability is not enough. A map can contain a valid path while still having very low optimal success probability, because slips can repeatedly push the agent into holes. In our Gymnasium random-map trial, many $p = 0.5$ and $p = 0.6$ maps produced near-zero oracle success and near-zero learned success for every algorithm. Such maps are useful as stress tests, but they are not useful for ranking algorithms because the dominant factor is map impossibility rather than learning quality.

For this reason, the final comparison benchmark uses carefully designed maps. The design goal is not to make the problem easy, but to make it comparable: each map should be stochastic, solvable, non-trivial, and reproducible. We therefore keep standard maps, especially the Gymnasium standard 8×8 map, for baseline comparability; add controlled 4×4 maps with exactly four holes; and use controlled 8×8 random maps for the main scalability comparison. The Gymnasium random-map results are retained to document why this design choice was necessary.

3.2 Environment

All reported experiments use `slippery=True`. In each step, the environment samples one of three outcomes: the intended direction, the left deviation, or the right deviation, each with probability $1/3$. This matches the standard FrozenLake stochastic transition idea and prevents the task from becoming deterministic maze following.

We evaluate four stochastic settings:

- **Standard 4x4:** the Gym-style map `SFFF/FHFH/FFFH/HFFG`, with four holes out of sixteen cells.
- **Designed 4x4:** random maps with exactly four holes, BFS reachability checks, and Value Iteration oracle filtering.
- **Fixed 8x8:** the standard fixed 8×8 benchmark map used in the project.
- **Gymnasium standard 8x8:** the library standard 8×8 map, used as a key standard comparison and sanity check.
- **Gymnasium random-map diagnostic:** generated Gymnasium maps at $p \in \{0.8, 0.7, 0.6, 0.5\}$, three maps per p . This trial motivated controlled-map design because many low- p maps had near-zero oracle success.
- **Controlled random 8x8:** random maps with $p \in \{0.80, 0.85, 0.90, 0.95\}$, five maps per difficulty. These maps are the main 8×8 algorithm-comparison benchmark.

The controlled 4×4 design prevents a different failure mode: unconstrained high- p random generation can create trivial maps with too few holes. For 8×8 , the controlled random maps are appropriate for comparing algorithms because they produce solvable stochastic tasks with meaningful variation. The Gymnasium maps are reported separately because several low- p maps have very low oracle success under slippery dynamics; mixing them directly with the controlled maps would make the mean score mainly reflect map impossibility rather than algorithm quality.

3.3 Algorithms and Metrics

The main comparison uses seven learned tabular methods: Q-learning, SARSA, Expected SARSA, SARSA(λ), Q(λ), Dyna-Q, and optimistic Q-learning. The exploration ablation additionally compares UCB-Q and softmax Q-learning. Value Iteration is used as an oracle, not as a fair model-free competitor, because it uses the known transition model.

The primary metric is greedy evaluation success rate after training. We also report standard deviation across seeds or maps and compare learned policies against the Value Iteration oracle. A high oracle score with a low learned score means the map is solvable but the learning algorithm failed to discover or propagate reward efficiently.

4 Results

4.1 Standard and Controlled 4x4

Figure 1 shows the latest standard 4×4 slippery rerun. The oracle reaches 0.847 success, which confirms that the stochastic setting is not expected to reach a guaranteed 1.0 success rate. Q-learning reaches 0.770, optimistic Q-learning reaches 0.767, SARSA reaches 0.723, and Expected SARSA reaches 0.680. These values are plausible for slippery FrozenLake and avoid the misleading 1.0 success rates from overly easy random maps.

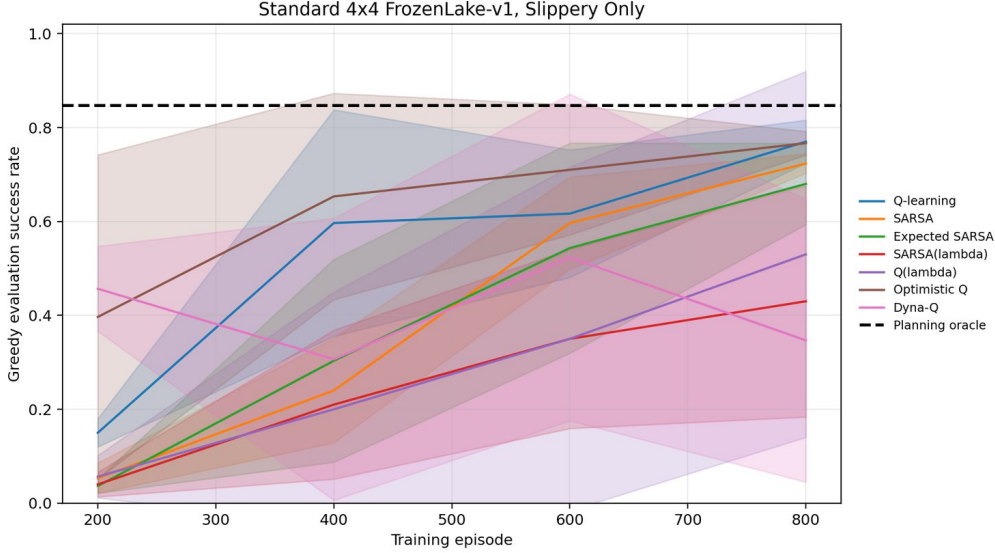


Figure 1: Standard 4×4 slippery FrozenLake results. The map has four holes, and all evaluation is stochastic.

Figure 2 reports the controlled random 4×4 experiment. These maps also have exactly four holes and are therefore much more credible than high- p random maps. With the longer 2,000-episode learning-curve rerun, Expected SARSA and Q-learning are the strongest mean final-success methods on these four maps, while optimistic Q-learning has the best mean AUC, meaning that it often learns earlier even when the final score is not always the highest.

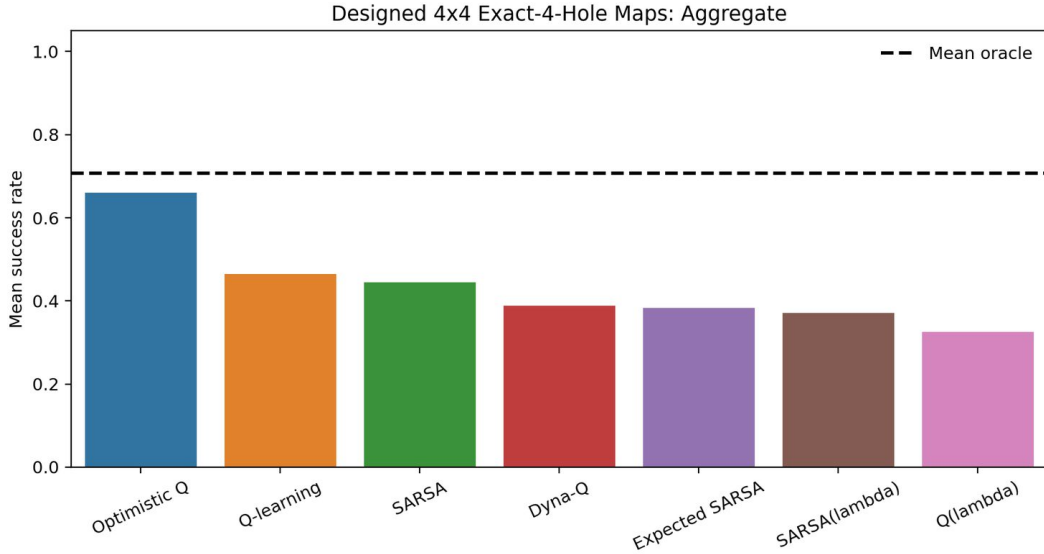


Figure 2: Designed random 4×4 maps with exactly four holes. This is the controlled random-map counterpart to the standard 4×4 map.

4.2 Controlled 8x8 Stochastic Benchmark

The controlled 8×8 setting is the main comparison benchmark in this report. It is more informative than 4×4 because sparse reward is harder to discover and the longer paths expose differences between exploration and reward-propagation strategies. At the same time, the maps are not arbitrary hard instances: they are designed to be solvable under slippery dynamics, so a low score is more likely to reflect the algorithm rather than an almost impossible layout.

Figure 3 summarizes the updated stochastic results. The random 8×8 numbers use 10,000 training episodes with evaluation every 500 episodes. Optimistic Q-learning is the strongest learned method on controlled random 8×8, reaching 0.988 mean final success over 20 maps. Expected SARSA, SARSA, Q-learning, $Q(\lambda)$, and Dyna-Q also improve substantially under the longer budget. These are the results that best support an algorithm-comparison story for the presentation and report.

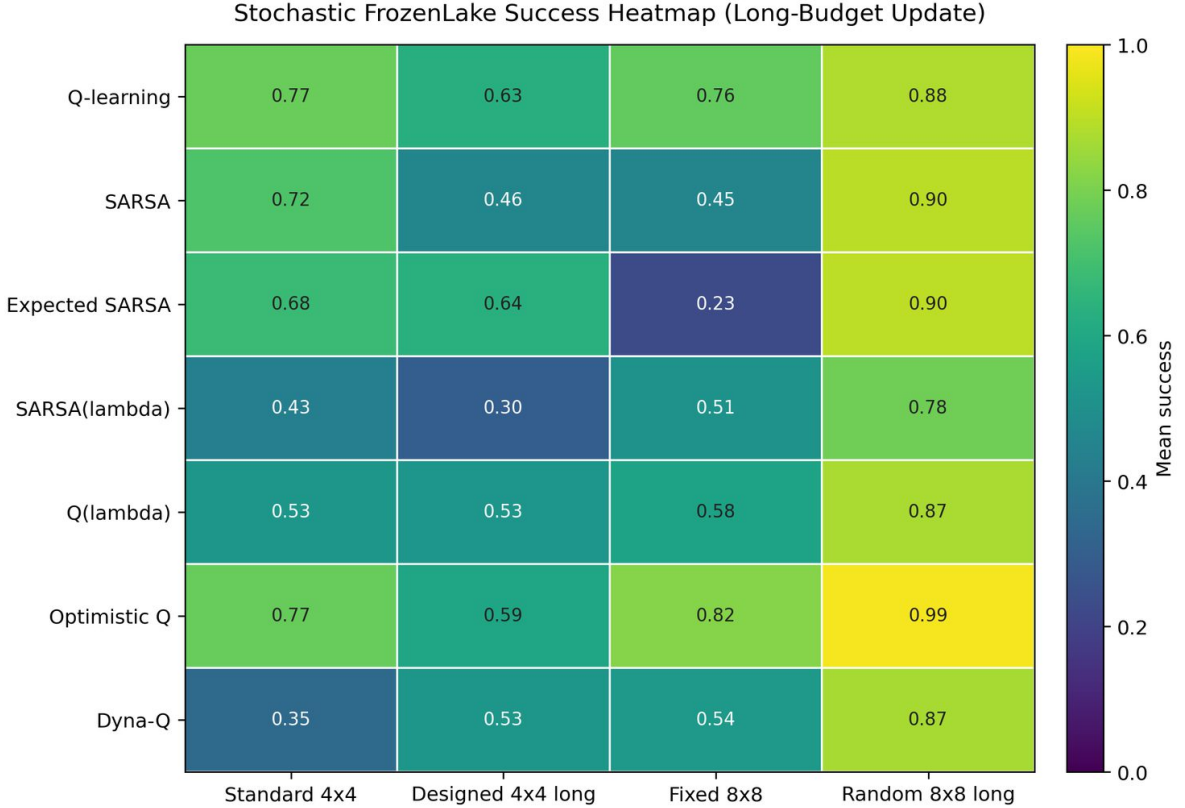


Figure 3: Mean success rate across the stochastic benchmark. The heatmap uses explicit per-cell text annotations and includes the long-budget random-map update.

Figure 4 shows the long-budget controlled random 8×8 difficulty sweep. Each point is an individual map, and the short horizontal marks indicate the mean for each algorithm and frozen-tile probability. This version is more interpretable than a single aggregate line plot because it exposes map-to-map variation directly. The high scores should therefore be interpreted as “solving the controlled stochastic benchmark”, not as solving every possible randomly generated 8×8 FrozenLake map.

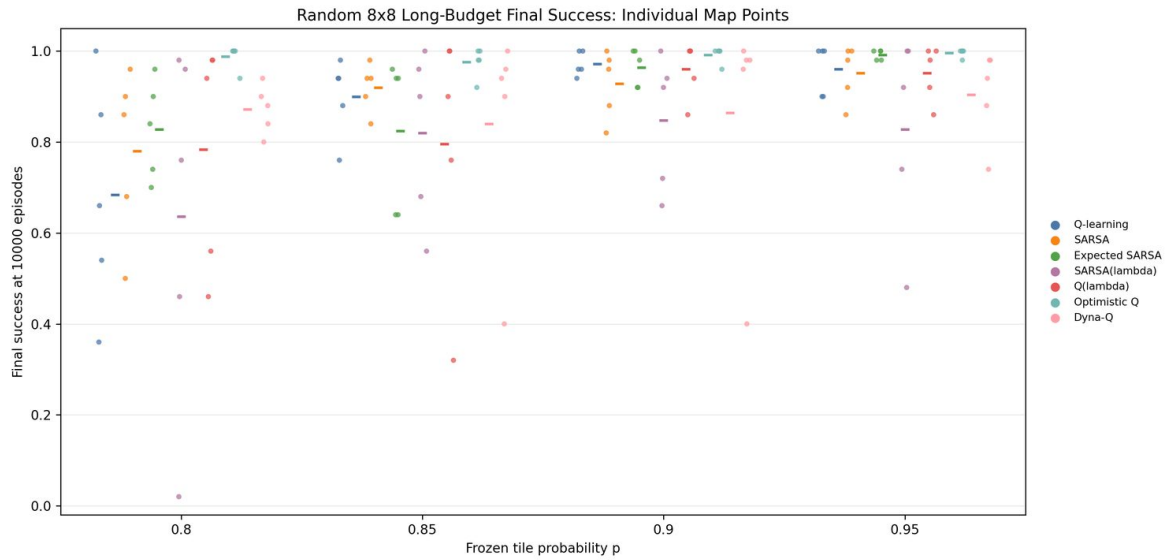


Figure 4: Random 8×8 slippery final success by frozen-tile probability after 10,000 training episodes. Each dot is one map.

Figure 5 shows per-map learning curves rather than one mixed curve. This directly addresses the variance problem: different maps have different learning difficulty even when their oracle success is high. The per-map plots show when an algorithm genuinely fails to converge on a specific layout.

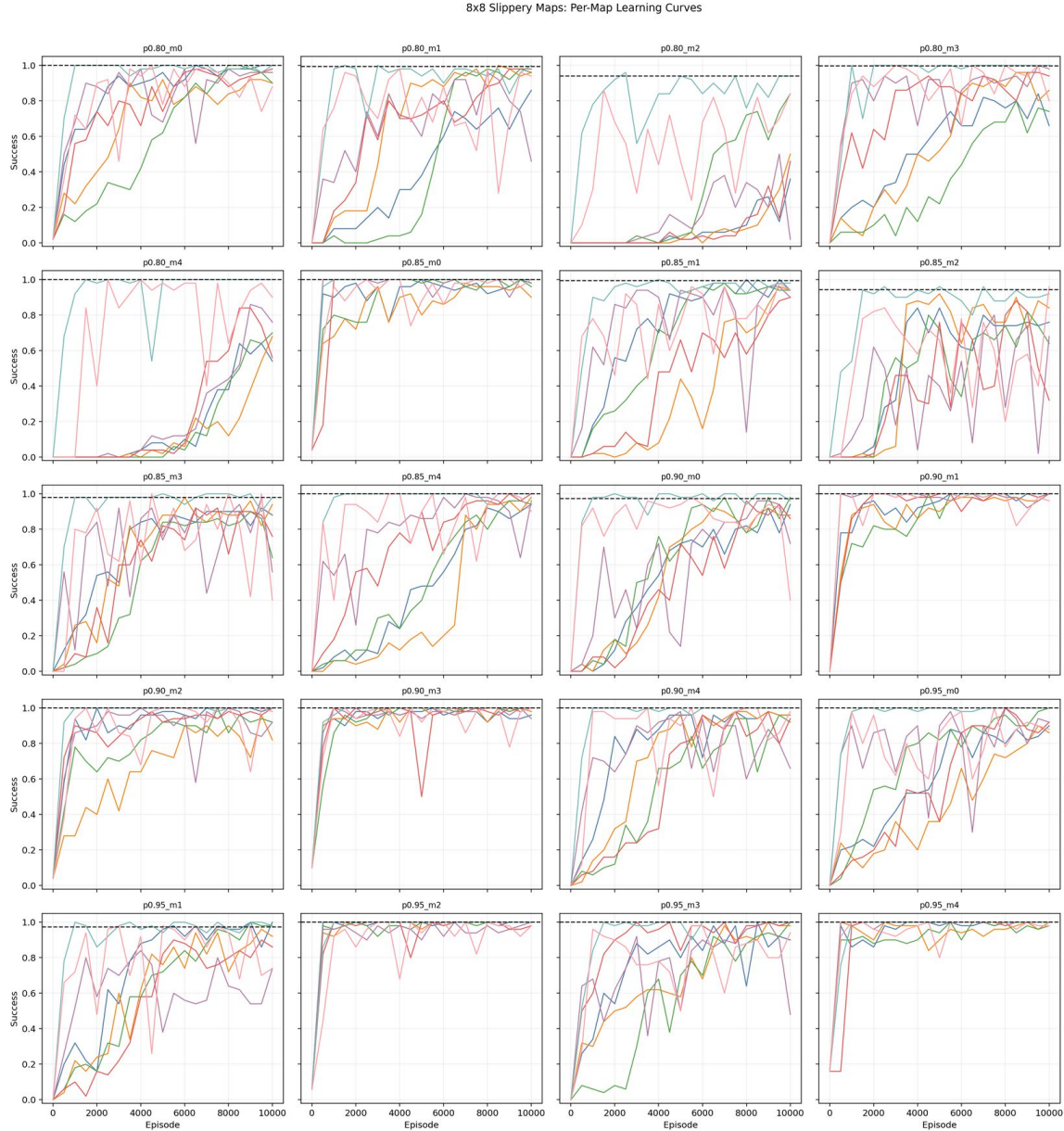


Figure 5: Per-map learning curves for 20 random 8×8 slippery maps. Each subplot contains all seven algorithms and the Value Iteration oracle.

4.3 Gymnasium Standard 8x8 and Random-Map Diagnostic

The Gymnasium 8×8 results have two different roles. First, the standard Gymnasium 8×8 map is an important benchmark. It is widely used, reproducible, and challenging without being degenerate. In our run, its oracle success is 0.861, and the learned mean success across the seven algorithms is 0.709. Several algorithms learn strong policies on this map: Expected SARSA reaches 0.85, $Q(\lambda)$ reaches 0.85, optimistic Q-learning reaches 0.82, Q-learning reaches 0.78, SARSA(λ) reaches 0.76, and SARSA reaches 0.74. This confirms that the training pipeline can handle a standard stochastic 8×8 benchmark.

Second, Gymnasium-generated random maps are used as a diagnostic trial. They should be read differently from the standard Gymnasium map and from the controlled benchmark above. They answer a design question: what happens if we rely directly on the library’s random-map generator at lower frozen-tile probabilities?

The answer is that performance drops sharply on many generated random maps, and the reason is visible from the oracle. Several $p = 0.5$ and $p = 0.6$ random maps have oracle success near zero under slippery dynamics. These maps are technically connected, because the generator guarantees a path from start to goal, but the stochastic transition model makes reliable completion extremely unlikely. Across all 13 Gymnasium maps, including the standard map and the random maps, optimistic Q-learning remains the best learned method, but its mean final success is only 0.235 because the near-impossible generated maps dominate the average. This result explains why the project moved toward carefully designed maps. It does not invalidate the standard Gymnasium 8×8 result or

the controlled-map results; it shows that random-map generation must be treated as an experimental variable.

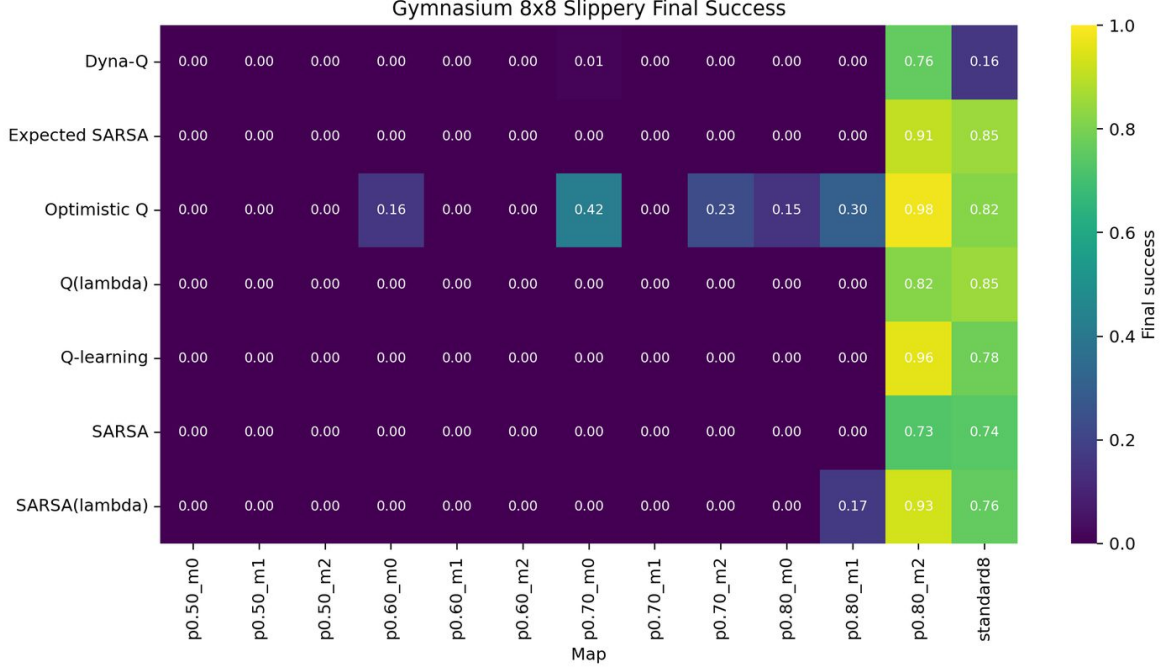


Figure 6: Gymnasium 8×8 slippery final success. The standard 8×8 map is a useful benchmark, while several low- p generated maps are too difficult for fair ranking.

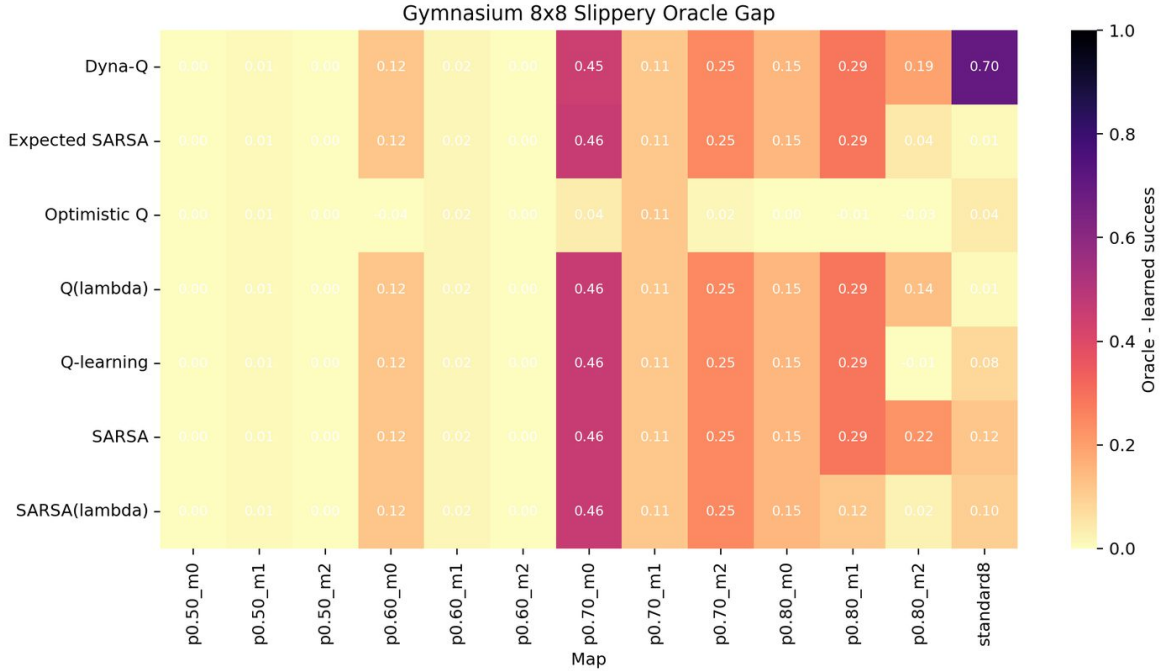


Figure 7: Oracle gap on Gymnasium 8×8 maps. The standard map is solvable, while several generated low- p maps are almost impossible under slippery dynamics.

4.4 Key Numerical Results

Table 1 reports the latest stochastic-only results used in the analysis. The controlled random 8×8 rows are the main random-map algorithm comparison; the Gymnasium rows are included separately as diagnostic evidence for why controlled maps were introduced.

Table 1: Selected latest stochastic results. Success values are mean final greedy evaluation rates.

Setting	Method	Mean Success	Std. Dev.
Standard 4×4	Value Iteration oracle	0.847	–
Standard 4×4	Q-learning	0.770	0.046
Standard 4×4	Optimistic Q-learning	0.767	0.025
Standard 4×4	SARSA	0.723	0.021
Designed 4×4	Value Iteration oracle	0.708	–
Designed 4×4	Expected SARSA	0.635	0.199
Designed 4×4	Q-learning	0.630	0.234
Designed 4×4	Optimistic Q-learning	0.590	0.308
Fixed 8×8	Optimistic Q-learning	0.817	0.155
Fixed 8×8	Q-learning	0.760	0.046
Fixed 8×8	Q(λ)	0.580	0.053
Random 8×8	Value Iteration oracle	0.988	–
Random 8×8	Optimistic Q-learning	0.988	0.023
Random 8×8	Expected SARSA	0.902	0.122
Random 8×8	SARSA	0.895	0.122
Gymnasium standard 8×8	Expected SARSA	0.850	–
Gymnasium standard 8×8	Q(λ)	0.850	–
Gymnasium standard 8×8	Optimistic Q-learning	0.820	–
Gymnasium all 8×8 maps	Optimistic Q-learning	0.235	0.327
Gymnasium all 8×8 maps	SARSA(λ)	0.143	0.317

5 Ablation Study

5.1 Exploration Ablation

Figure 8 isolates exploration strategy on fixed 8×8 slippery FrozenLake. Optimistic Q-learning reaches 0.817 and Q-learning reaches 0.760. UCB-Q and softmax Q-learning perform poorly under this training budget, reaching 0.133 and 0.100. The result suggests that simple optimistic initialization is especially effective in this sparse tabular task: unknown actions remain attractive long enough for the agent to discover the goal.

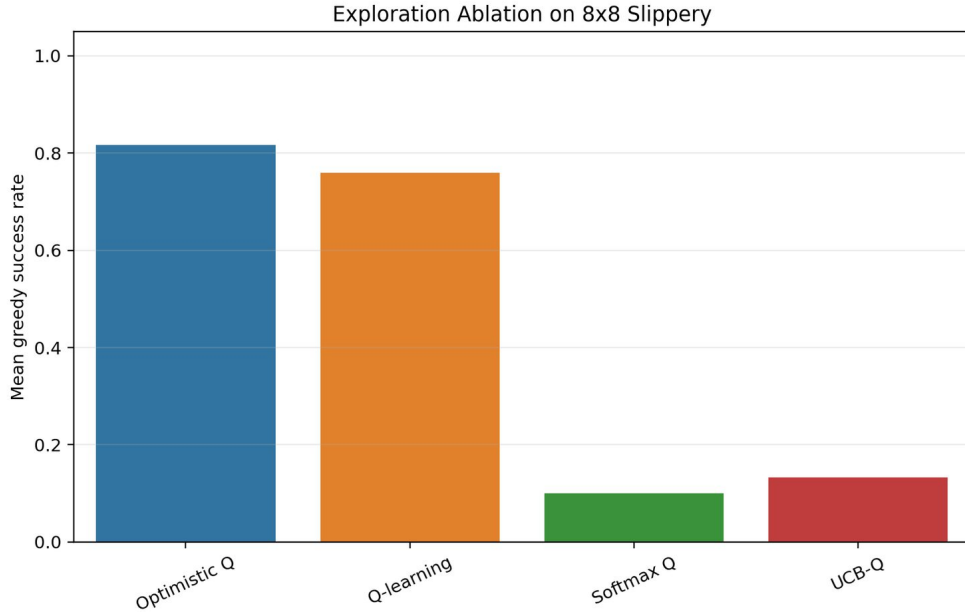


Figure 8: Exploration ablation on fixed 8×8 slippery FrozenLake. Optimistic initialization is the strongest exploration mechanism here.

5.2 Reward Propagation Ablation

Figure 9 compares methods that propagate sparse terminal reward in different ways. Q-learning is strongest on the fixed 8×8 map at 0.760. $Q(\lambda)$, Dyna-Q, and SARSA(λ) are lower on this fixed map but remain important because they improve robustness on random 8×8 maps. n-step SARSA and Expected SARSA are highly variable, which is expected when successful episodes are rare and the behavior policy is still exploratory.

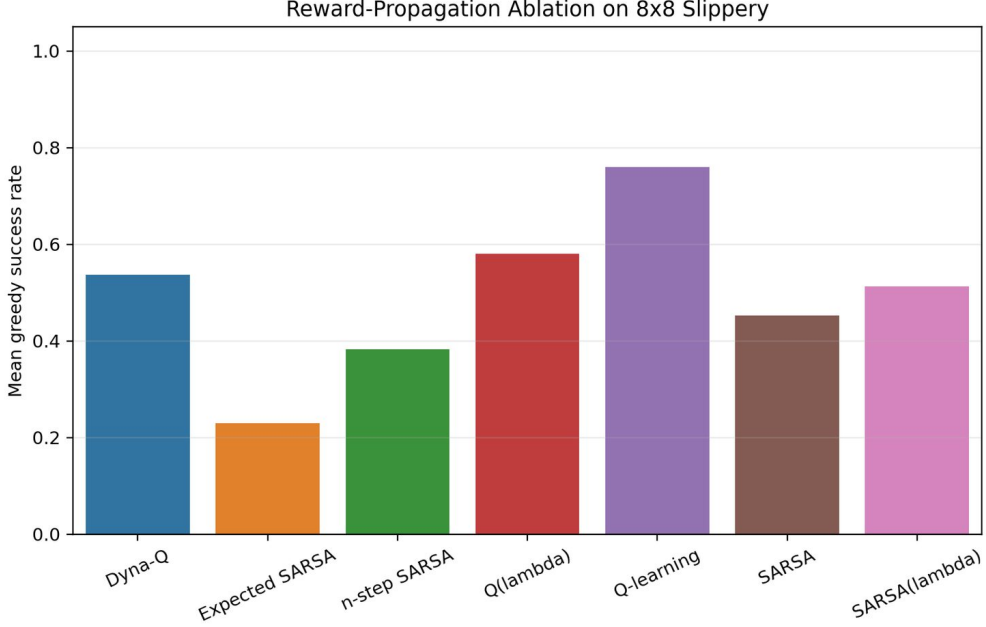


Figure 9: Reward-propagation ablation on fixed 8×8 slippery FrozenLake. More complex propagation is not automatically better under short sparse-reward training.

5.3 Oracle-Gap Ablation

Figure 10 measures how far learned policies are from the known-model oracle on each random 8×8 map. Optimistic Q-learning has the smallest mean gap after the long run. Some non-optimistic methods still have isolated hard-map failures, which is why per-map heatmaps are more useful than a single averaged line.

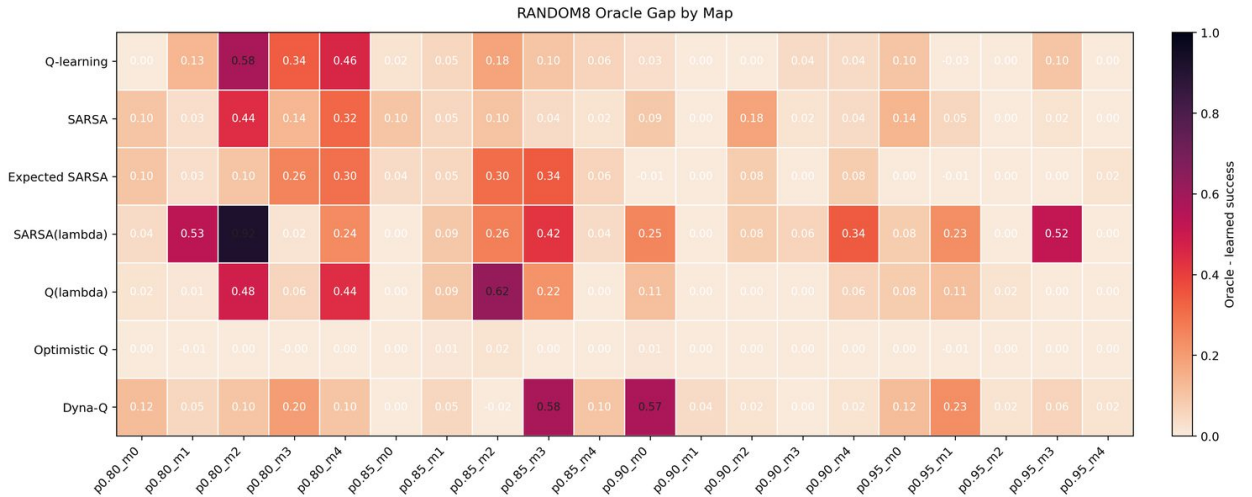


Figure 10: Oracle gap on random 8×8 slippery maps after 10,000 training episodes. Lower is better. Every cell is explicitly annotated.

Figure 11 gives two additional views: AUC measures how good the whole learning process is, while the convergence heatmap reports the first episode at which success reaches 0.80. Dyna-Q has a high mean AUC and early average convergence, even though its final score is slightly below the strongest methods.

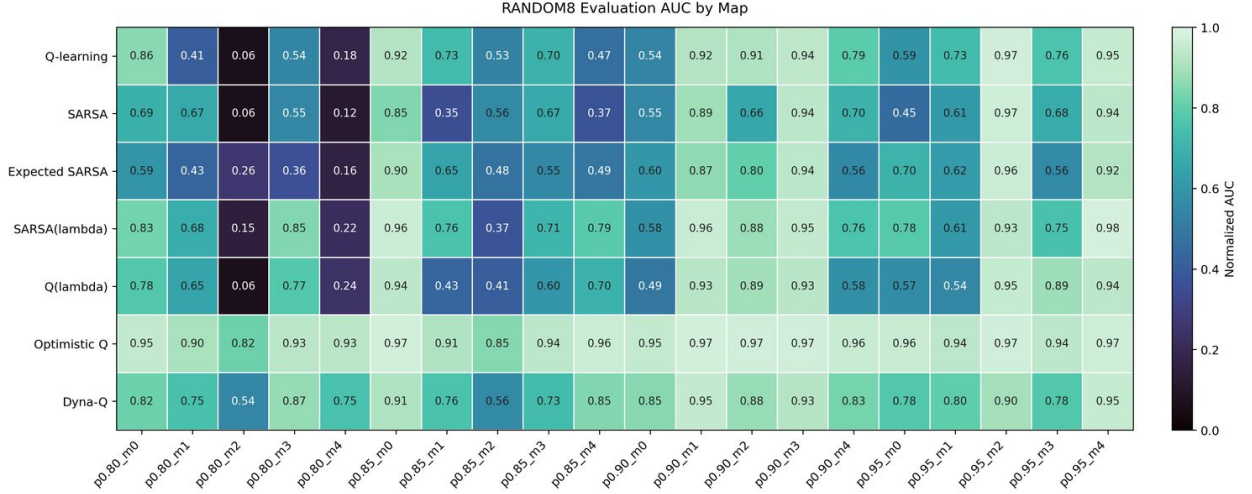


Figure 11: Evaluation AUC by algorithm and random 8×8 map. Higher values mean better performance throughout training, not only at the end.

6 Hyperparameter Optimization

We use Optuna with the TPE sampler because it is simple to run locally, supports conditional search spaces, and is sufficient for the size of this project. A key lesson is that the search space must match the algorithm. A generic search over learning rate, discount factor, and epsilon schedule is incomplete for methods like optimistic Q-learning and Dyna-Q. After adding `optimistic_init` and `planning_steps`, tuning became more meaningful. On fixed 8×8 stochastic FrozenLake, the best Optuna validation scores were 0.830 for optimistic Q-learning, 0.795 for Dyna-Q, 0.735 for Q-learning, and 0.665 for Expected SARSA. Three-seed validation confirmed tuned optimistic Q-learning and tuned Dyna-Q as the strongest tuned candidates. Optuna is therefore a good choice for this report; Ray Tune would be more appropriate only if we scaled to many more maps or GPU-based neural RL.

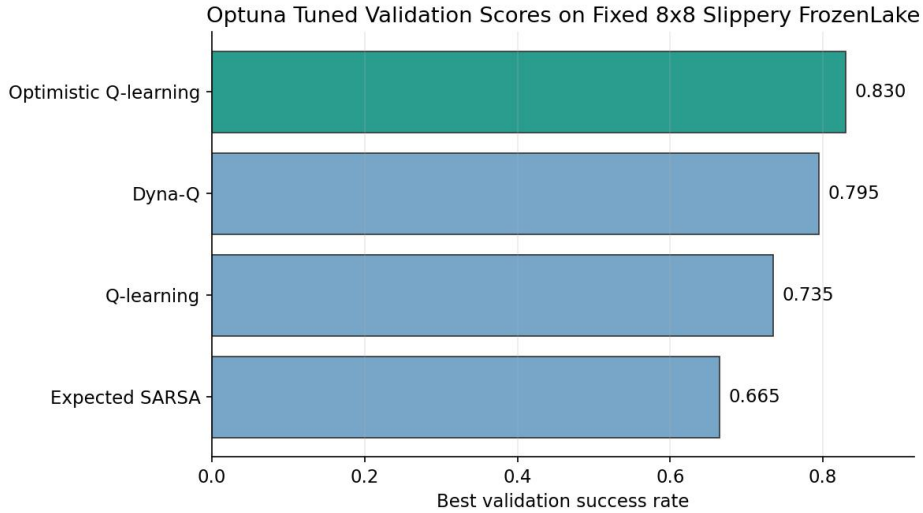


Figure 12: Best Optuna validation scores on fixed 8×8 stochastic FrozenLake.

7 Discussion

The main conclusion is that stochastic FrozenLake rewards exploration more than raw algorithmic complexity, but only after the benchmark maps are defined carefully. Optimistic Q-learning performs best on the controlled 8×8 benchmark because optimistic initial values actively encourage the agent to try uncertain actions, which is crucial when the only positive reward is at the goal. The long-budget results also clarify an earlier concern: the large standard deviations in short-budget random-map plots mainly came from mixing maps with different layouts before the algorithms had enough time to converge. When each map is plotted separately and the 8×8 budget is

increased to 10,000 episodes, most tabular methods improve greatly on the controlled maps.

Dyna-Q and eligibility traces answer a different problem: once reward is found, how quickly can it be reused or propagated backward? In the final-success ranking, Dyna-Q is not the best long-budget method, but its AUC and convergence behavior are strong. This means Dyna-Q is valuable when sample efficiency during training matters, while optimistic Q-learning remains the strongest final-policy choice.

The map design results are important. A single fixed map is reproducible but not enough for validation, but some fixed maps are still essential standards. The Gymnasium standard 8×8 map is one such case: it is a useful common reference and confirms that our algorithms can learn a recognized stochastic benchmark. Random maps are also valuable, but their role must be stated clearly. Our workflow reflects this: the Gymnasium random-map attempt exposed many near-zero oracle maps, so we moved to controlled maps for the main comparison. For 4×4 , maps need a controlled hole count, reachability checks, and oracle filtering; otherwise the benchmark can accidentally become too easy. For 8×8 , the controlled random maps are appropriate for comparing algorithms because they produce solvable stochastic tasks with meaningful variation. The Gymnasium random-map diagnostic remains valuable because it explains the boundary of the claim: lower- p random maps may be connected in the deterministic sense but nearly impossible under slippery transitions.

We intentionally exclude exploratory neural-RL stress tests from the final benchmark because their current results were not competitive on this sparse tabular task under the available budget. Including them would distract from the validated stochastic tabular comparison. Similarly, 16×16 maps are kept as future work because current results depend strongly on special map design, reward shaping, or much larger training budgets.

8 Conclusion

We converted the original 4×4 Q-learning project into a stochastic FrozenLake benchmark with multiple tabular algorithms, controlled random maps, Value Iteration oracle references, Optuna tuning, ablation studies, and long-budget per-map learning curves. The key methodological contribution is the benchmark-design refinement: the Gymnasium standard 8×8 map provides an important standard comparison, but direct Gymnasium random maps were too often near-impossible under slippery dynamics, so controlled maps were introduced to make algorithm comparison meaningful. The central algorithmic result is that optimistic Q-learning is the strongest learned method on the controlled stochastic benchmark, achieving near-oracle performance on controlled random 8×8 maps. Expected SARSA, SARSA, and Q-learning form a strong second group, while Dyna-Q is more attractive when learning speed and AUC matter. The final practical recommendation is therefore: use Q-learning as a baseline, add optimistic exploration, validate on standard maps such as Gymnasium 8×8 , compare algorithms on controlled stochastic maps, and use Gymnasium-generated maps plus oracle gaps as a diagnostic check before making broad claims about random-map generalization.

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